CrisisMap

ML Inference Report

SiamUnet building damage assessment inference for Beirut Port

Assessment ID	ASM-D06926
Mode	Local inference, not training
Model	SiamUnet:model_best.pth.tar
Device	CPU
Study Area	Beirut Port, Lebanon

Key clarification: this document describes inference using an existing pretrained checkpoint. No local training or fine-tuning was executed in the previous run.

1. What Was Actually Run

The previous CrisisMap run did not train a new machine learning model. It loaded an existing pretrained checkpoint and used it for inference against Beirut pre/post Maxar imagery and building footprints.

- No local epochs were run.
- No local train/validation split was created for this execution.
- No local optimizer, training batch size, or loss curve exists for this run.
- The run produced inference predictions, evidence chips, validation metrics, and dashboard/report artifacts.

2. Code Map

Model architecture	data/raw/models/microsoft_building_damage_assessment_cnn_siamese/end_to_end_Siam_UNet.py
Inference wrapper	backend/app/services/geospatial/ml_inference.py
Beirut pipeline	backend/app/services/geospatial/imagery_baseline.py
Quality/chip API	backend/app/api/v1/routes/assessments.py
Priority evidence UI	frontend/src/app/priority/[id]/page.tsx

3. Architecture

The model is a Siamese UNet. It accepts a pre-disaster image chip and a post-disaster image chip. The shared UNet branch predicts building segmentation masks, while the difference between pre/post feature embeddings feeds a damage classification head.

Architecture	SiamUnet
Input A	Pre-disaster RGB chip, 256x256
Input B	Post-disaster RGB chip, 256x256
Segmentation outputs	Pre-event and post-event building mask logits
Damage output	5-class damage logits
Classes	background, no-damage, minor-damage, major-damage, destroyed

4. Pipeline Flow

- User AOI polygon limits the Beirut analysis area.
- Maxar building footprints are clipped to the AOI.
- Pre/post Maxar GeoTIFF windows are read per footprint.
- A 256x256 pre/post tensor pair is created per building.
- SiamUnet predicts pixel-level damage logits.
- Softmax probabilities are averaged into a building-level prediction.
- The ML score is blended with an image-change baseline score.
- Predictions are joined back into GeoJSON building polygons.
- Results are validated against Copernicus EMSR452 ground-truth points.
- Before/after chip evidence is exposed in Priority Detail.

5. Local Inference Configuration

```
ML_MODEL_ENABLED=true
ML_MODEL_CHECKPOINT_PATH=../data/raw/models/microsoft_building_damage_assessment_cnn_siamese/model_best.pth.tar
ML_MODEL_DEFINITION_PATH=../data/raw/models/microsoft_building_damage_assessment_cnn_siamese/end_to_end_Siam_UNet.py
ML_MODEL_DEVICE=cpu
torch==2.5.1+cpu
```

6. Result Summary

Pipeline method	ml-inference
ML inference count	60
Baseline fallback count	0
Buildings assessed	60
Runtime	~51.21 seconds on CPU
Matched validation buildings	54
Accuracy	0.778
Macro F1	0.221

7. Validation Charts

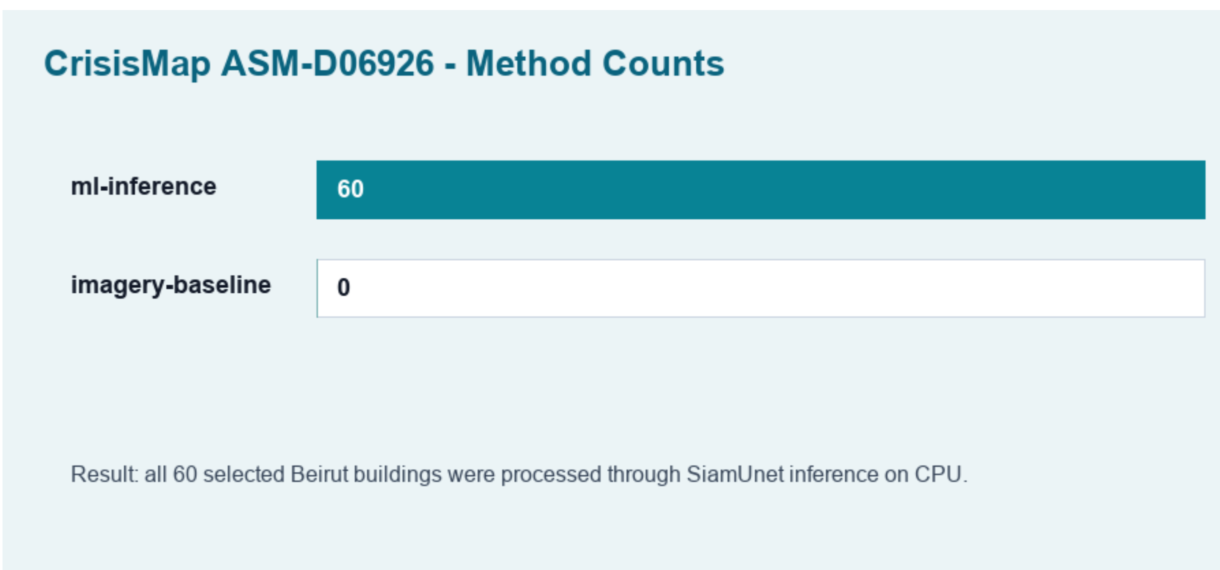

Figure 1. All 60 selected Beirut buildings were processed by ML inference.

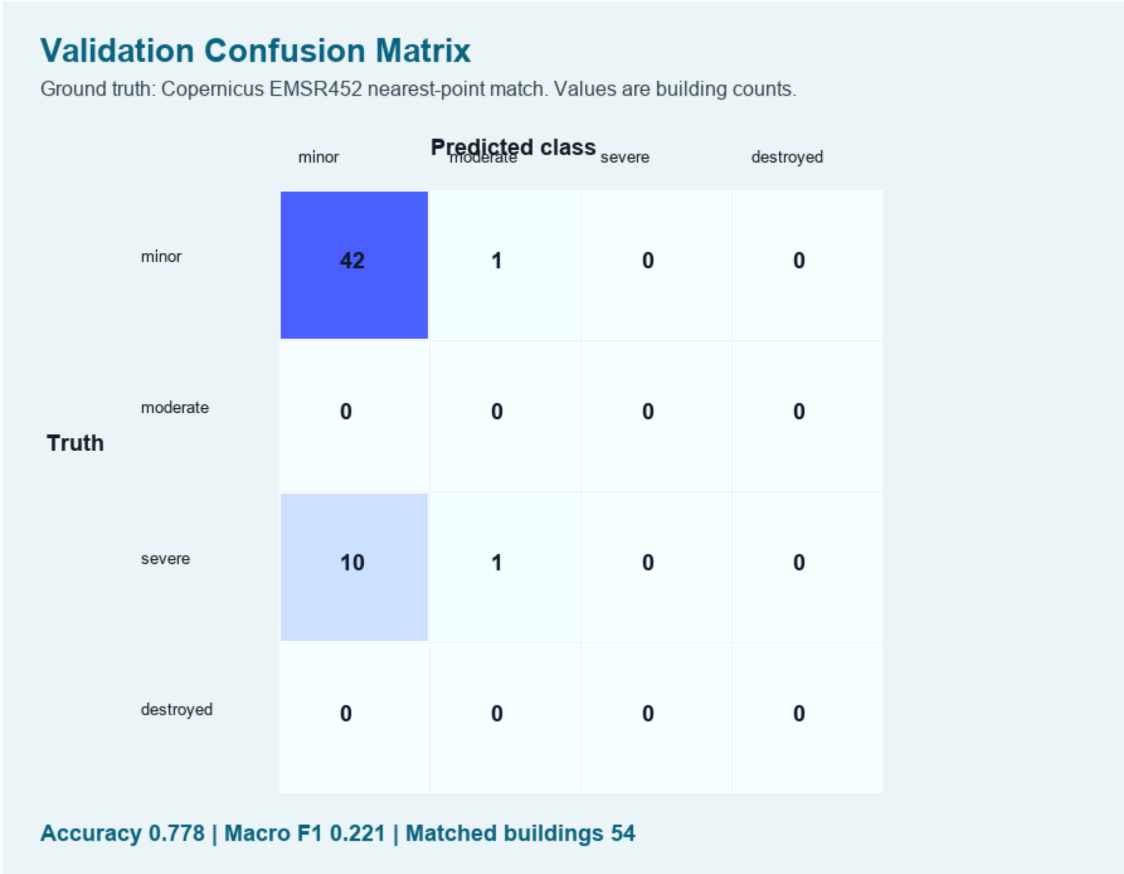

Figure 2. Confusion matrix against Copernicus EMSR452 nearest-point validation labels.

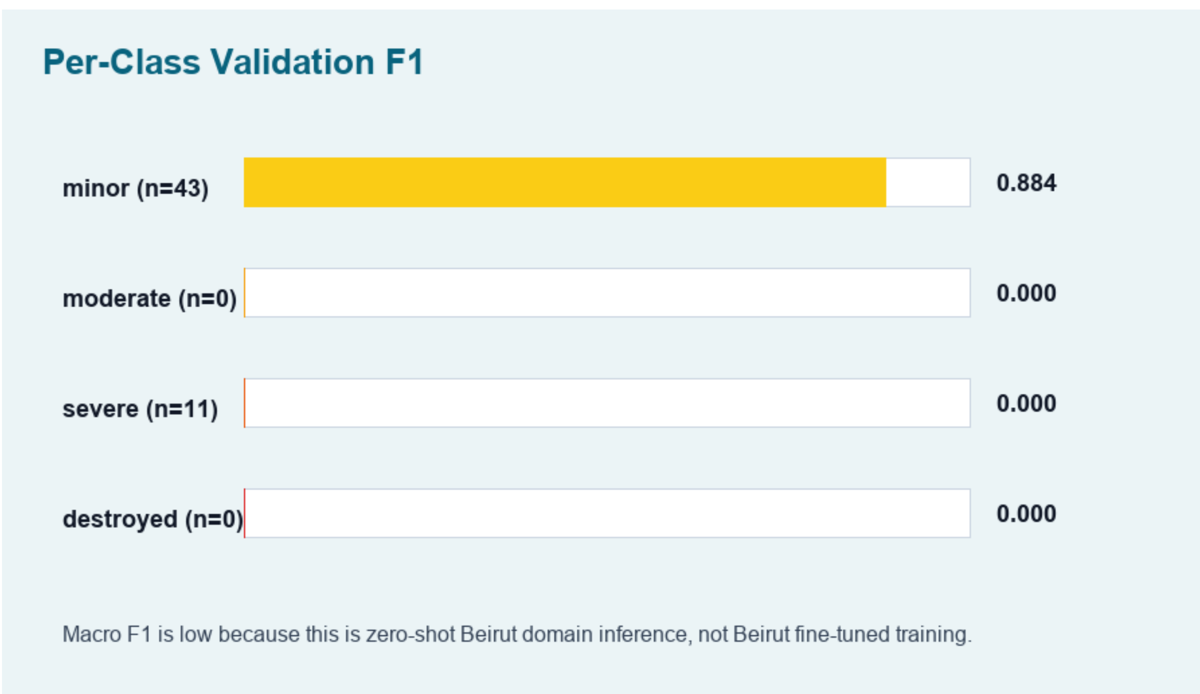

Figure 3. Per-class F1 score. Macro F1 remains low because this is zero-shot Beirut inference.

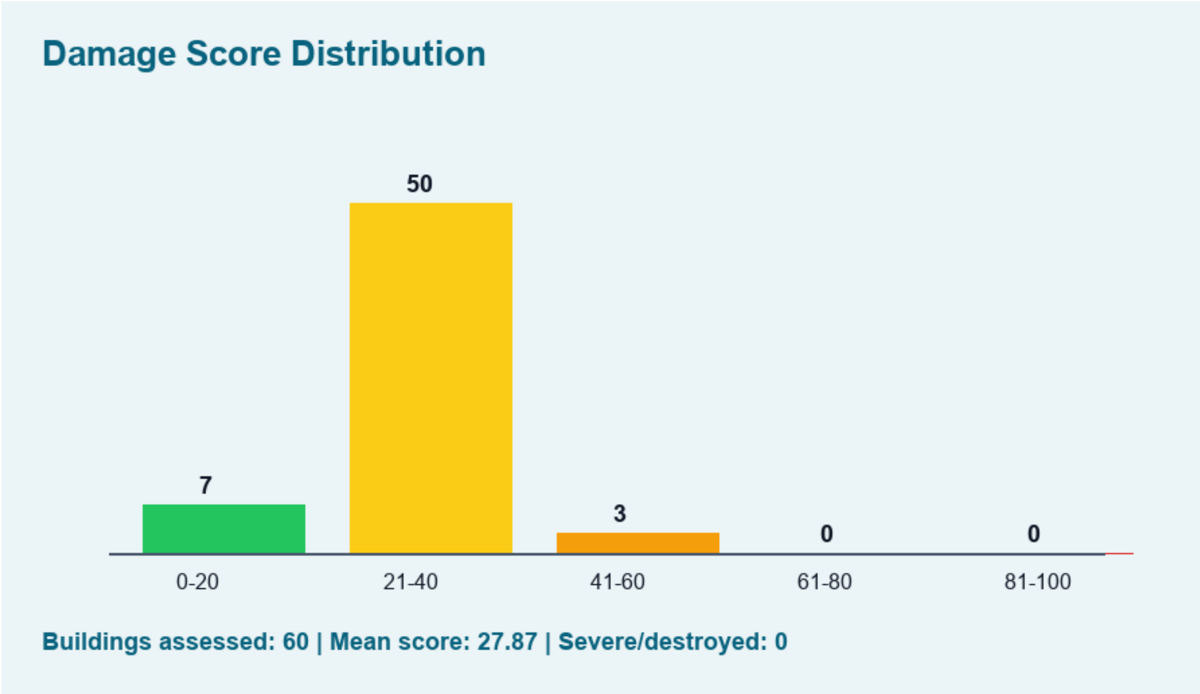

Figure 4. Distribution of final building-level damage scores.

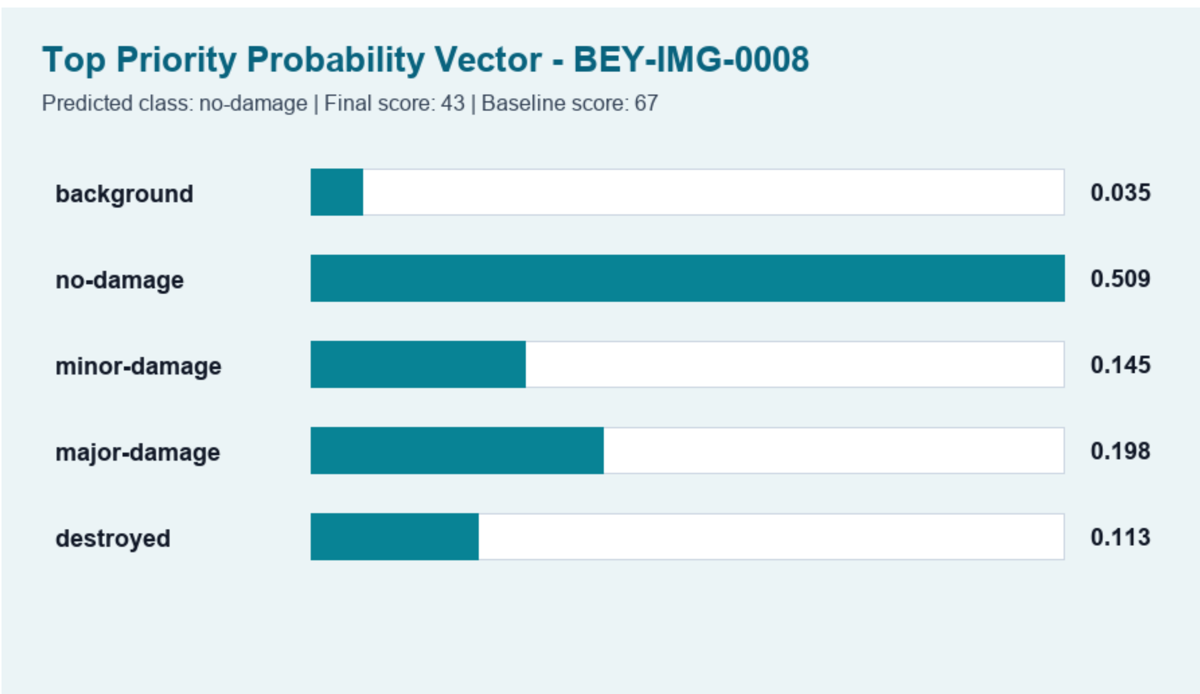

Figure 5. Probability vector for the top priority building.

8. Interpretation

The model currently detects many matched validation samples as minor/no-damage and under-detects severe damage in this Beirut sample. That is expected for zero-shot domain transfer from xBD-style disaster tiles to Beirut Maxar imagery and EMSR452 point labels.

- The pipeline is operational and can run real imagery inference locally.

- The validation score is not yet acceptable as final operational accuracy.
- Fine-tuning or threshold calibration is required before making strong claims about Beirut accuracy.

9. Recommended Next ML Step

- Create a Kaggle/Colab notebook for fine-tuning with xBD Challenge training set and Tier3 data.
- Use train/validation/test splits from xBD or a reproducible disaster-wise split to avoid leakage.
- Track loss, segmentation F1/IoU, damage macro F1, confusion matrix, and per-class recall.
- Export a new checkpoint and point ML_MODEL_CHECKPOINT_PATH to the fine-tuned model.
- Re-run Beirut validation and compare against the current baseline report.